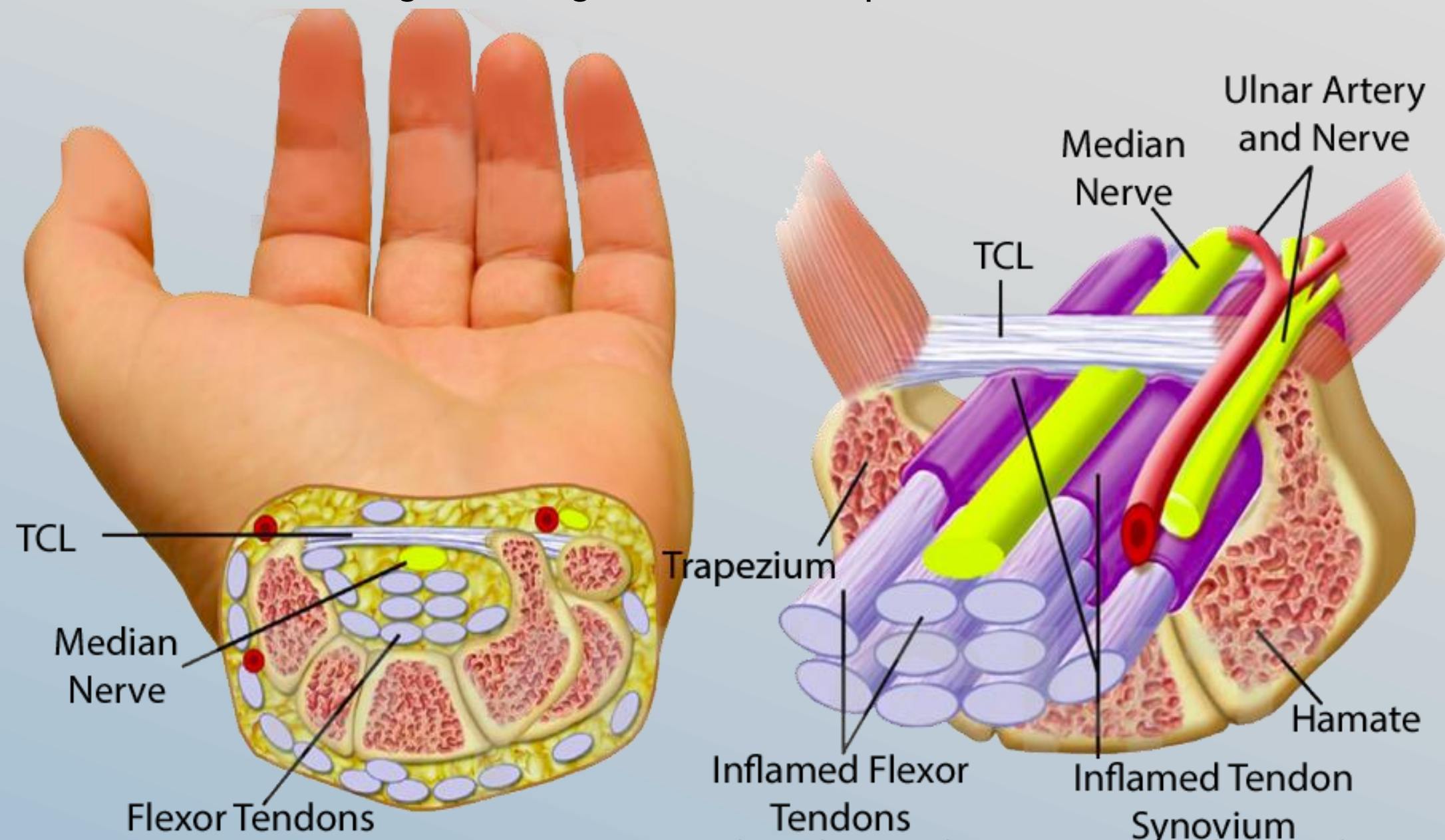


Project Goal

To provide users an effective, easy-to-use, and pleasurable treatment option which alleviates carpal tunnel syndrome symptoms and avoids more costly and invasive procedures. The AUTO-Trac is designed to stretch and warm the transverse carpal ligament and associated connective tissues around the carpal tunnel. This relieves pressure on the median nerve alleviating symptoms.

Carpal Tunnel Syndrome

Fig. 1 - Diagram of the Carpal Tunnel



Carpal Tunnel Syndrome (CTS) is an impingement of the median nerve in the carpal tunnel causing pain, numbness, tingling, and cold sensation. CTS affects approximately 3.7 million Americans annually.

Therapy Background

The C-TRAC is a manual traction device which uses an inflatable bladder and flexible frame to apply a 3-point bending force upon the transverse carpal ligament (TCL). This increases the cross sectional area of the carpal tunnel, relieving pressure on the median nerve.



Fig. 2 - C TRAC

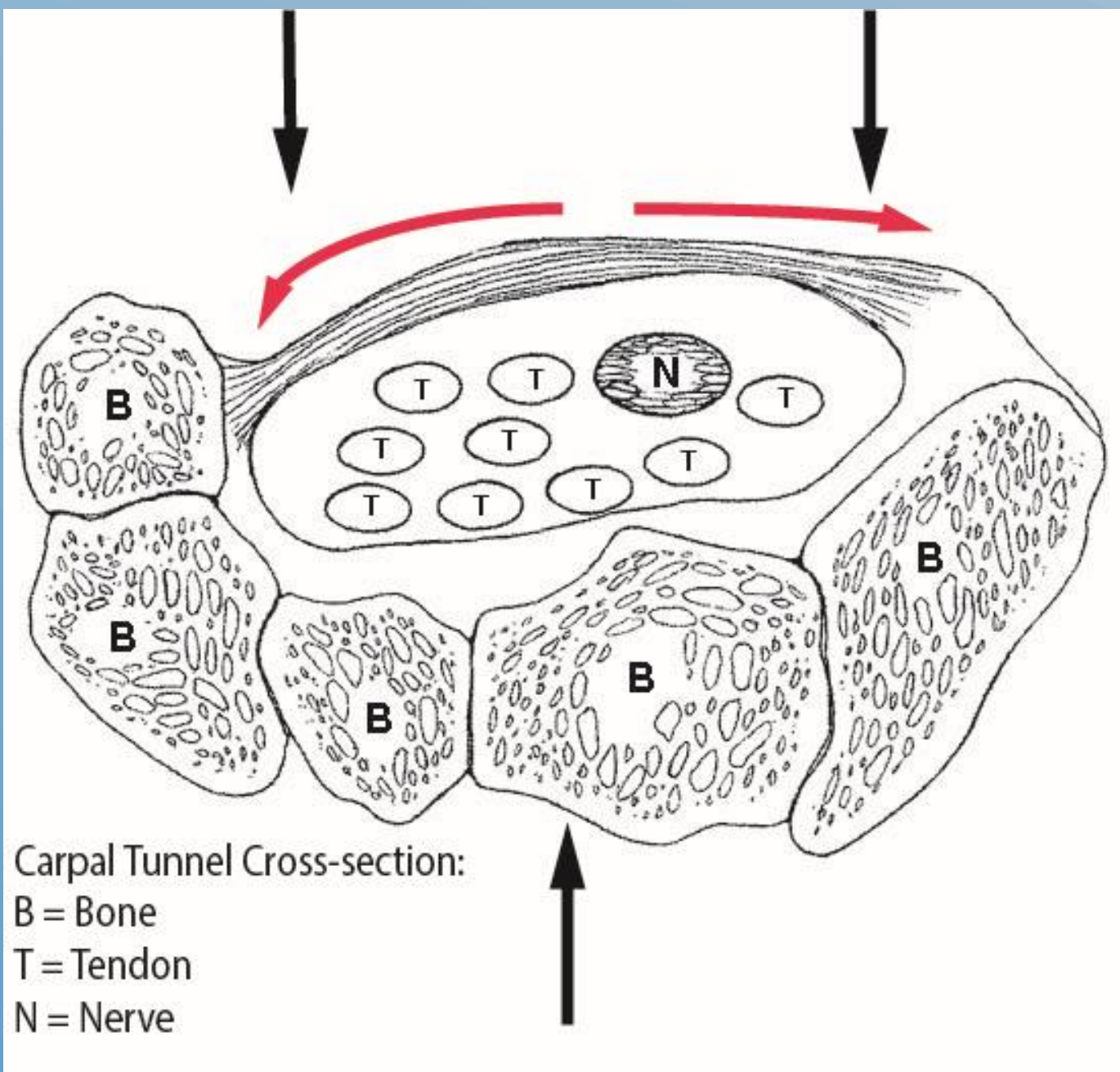


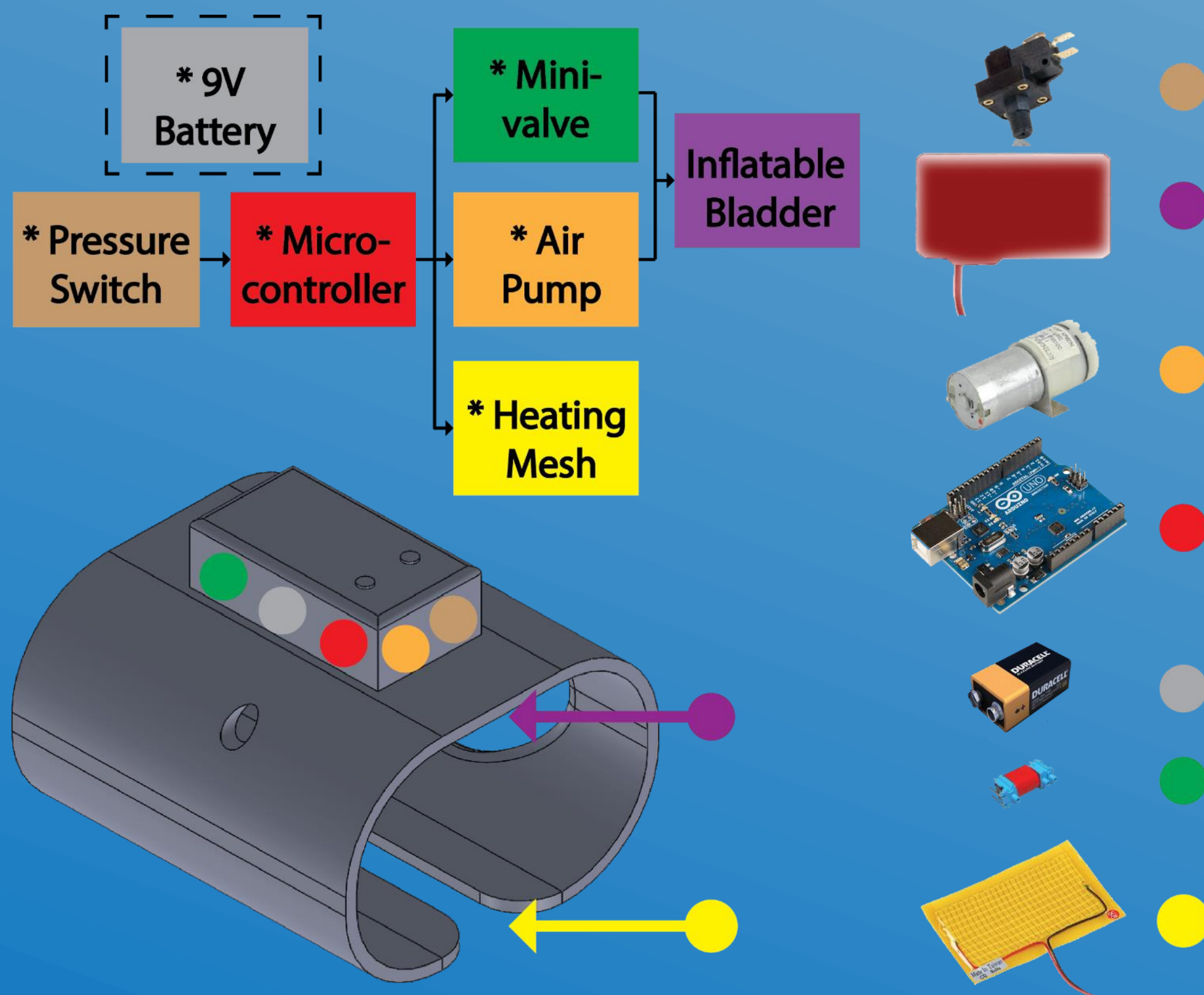
Fig. 3 - Cross Section of Wrist with 3-point force

A study on C-TRAC daily use after four weeks showed that patients had significant reduction or elimination of CTS symptoms.

Device Design

Our design improves the C-TRAC procedure:

- Automatic
- Applies Heat to patient's hand and wrist
- Easy to Use
- Intuitive



Modeling, Testing, and Results

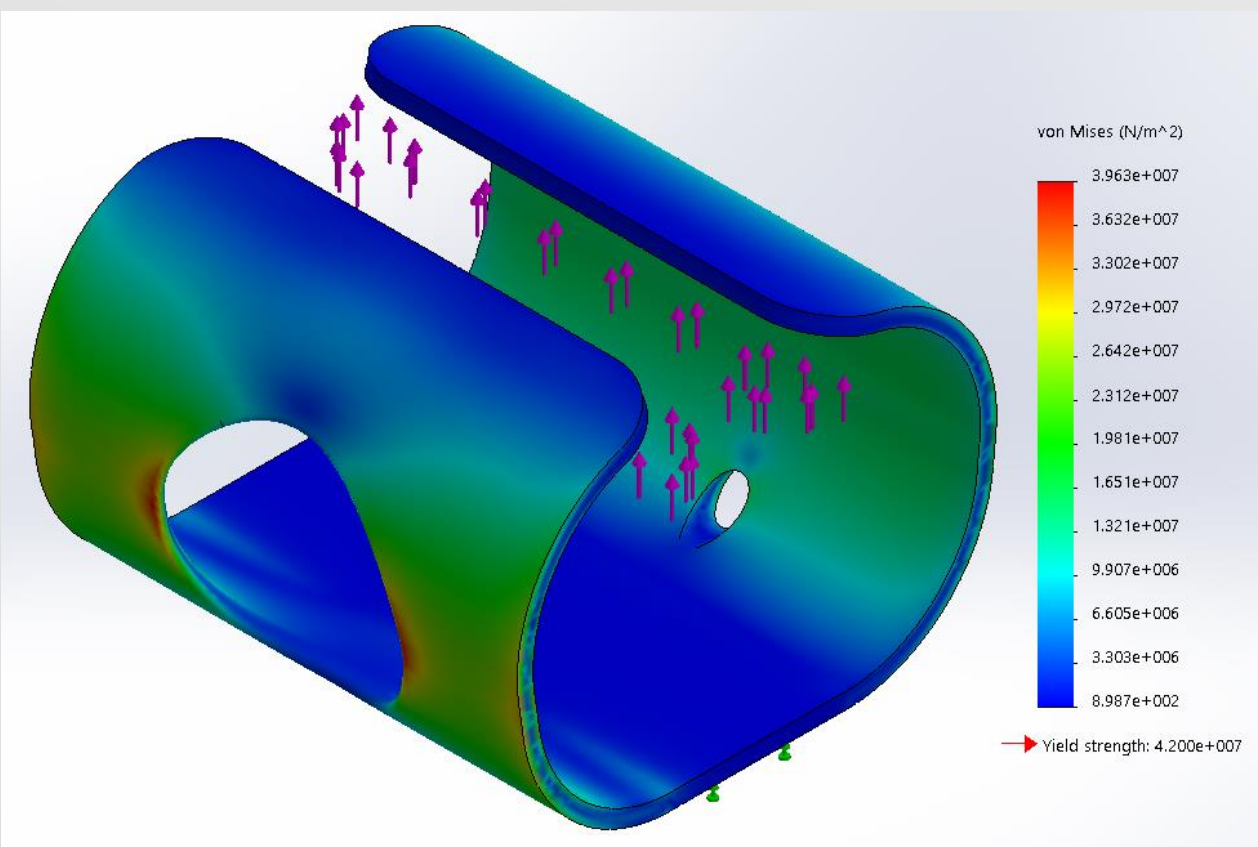


Fig. 3 - Solidworks FEA (Isometric Stress View)

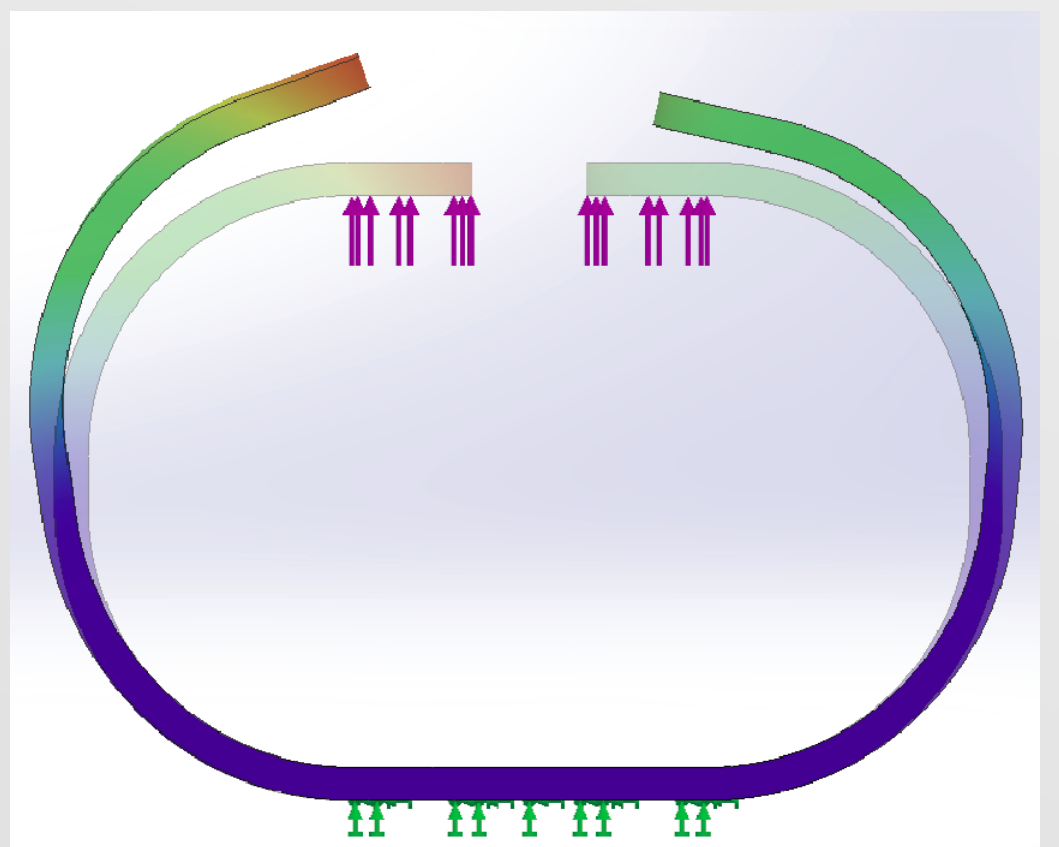


Fig. 4 - Solidworks FEA (Displacement View)

- Finite Element Analysis (FEA) in SolidWorks showed maximum stress along the long axis of the ovalar thumb opening
- Validated by Tensile Test on 3D printed frame
- Component Testing concluded timed control of Pressure and Heat
- Pressure Sensor terminates pumping and releases valve if pressure exceeds safety limit
- On/Off Button can terminate cycle and relieve pressure/heat at any time

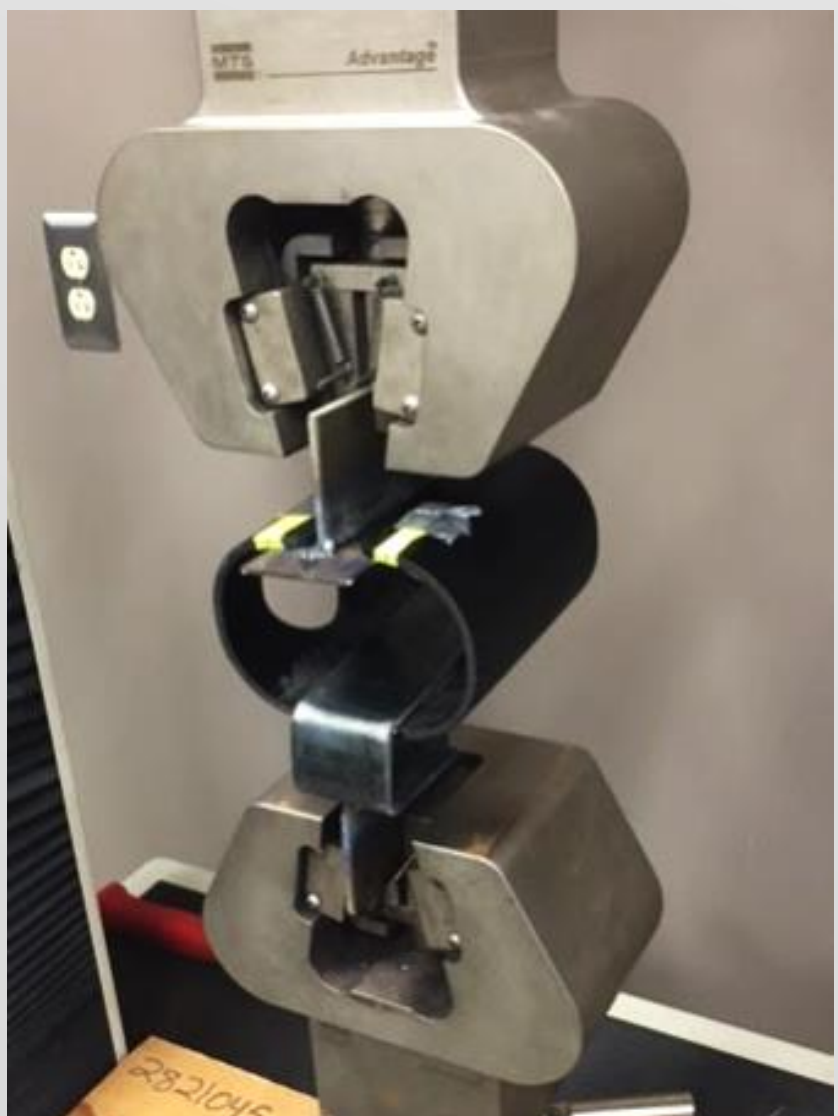


Fig. 5 - 3D Printed Frame Tensile Test

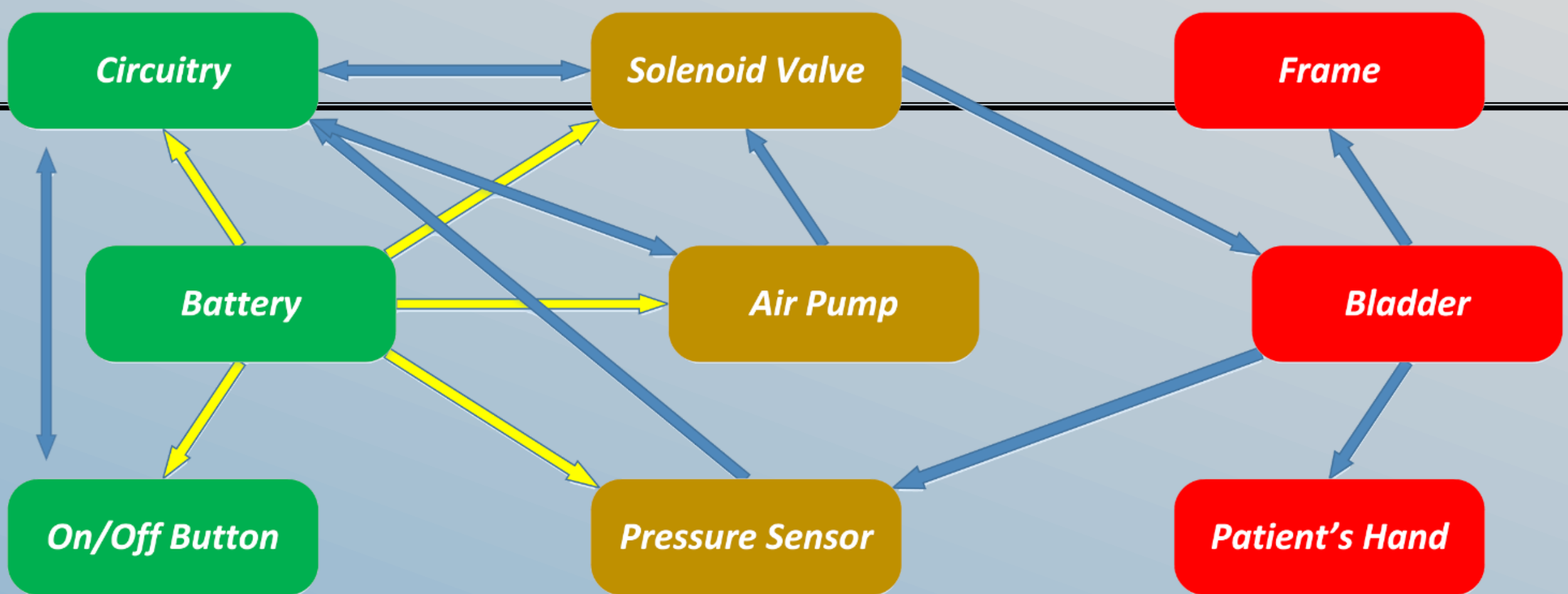


Fig. 6 - AUTO-Trac Block Diagram

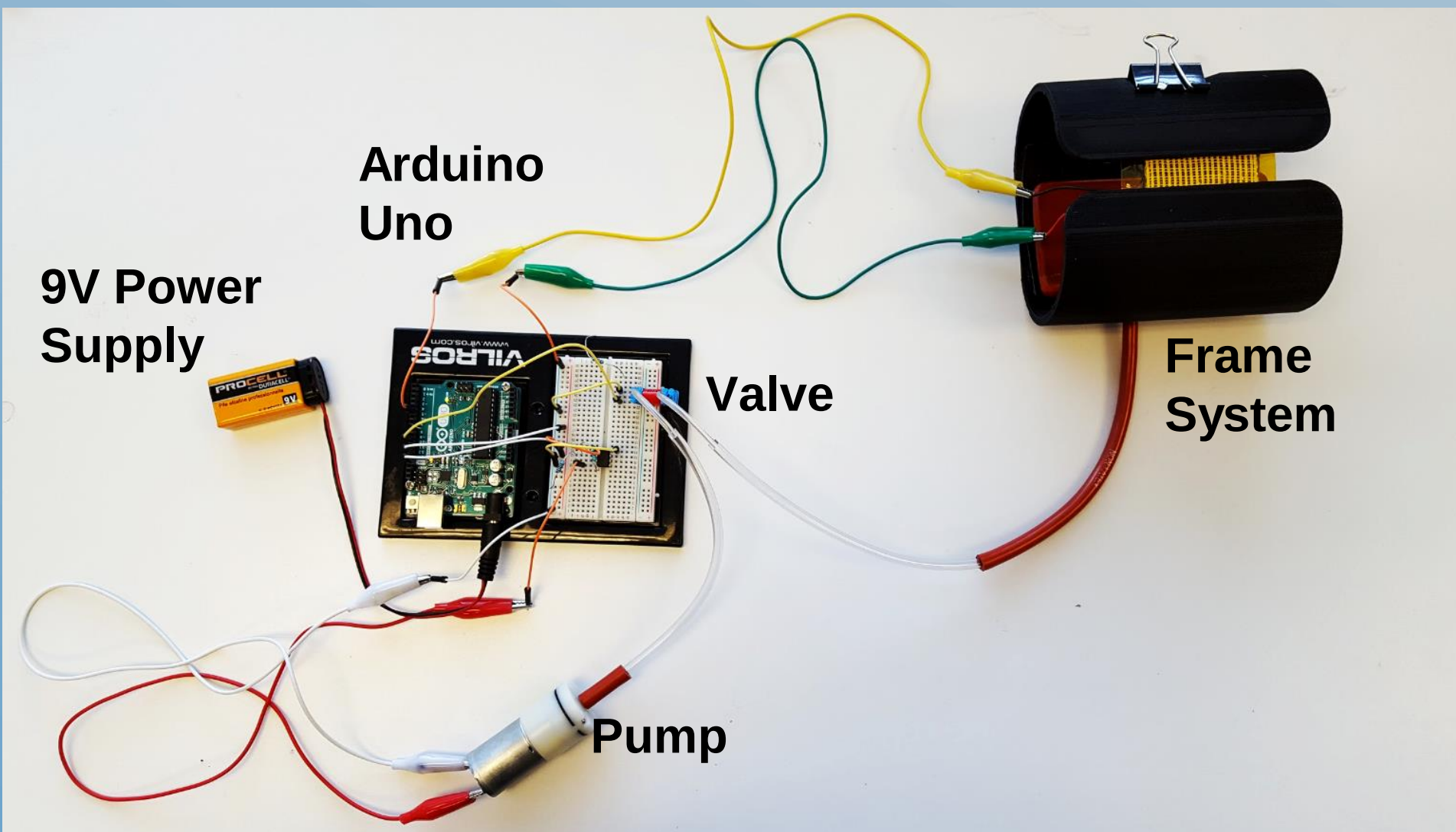


Fig. 7 - Total AUTO-Trac System

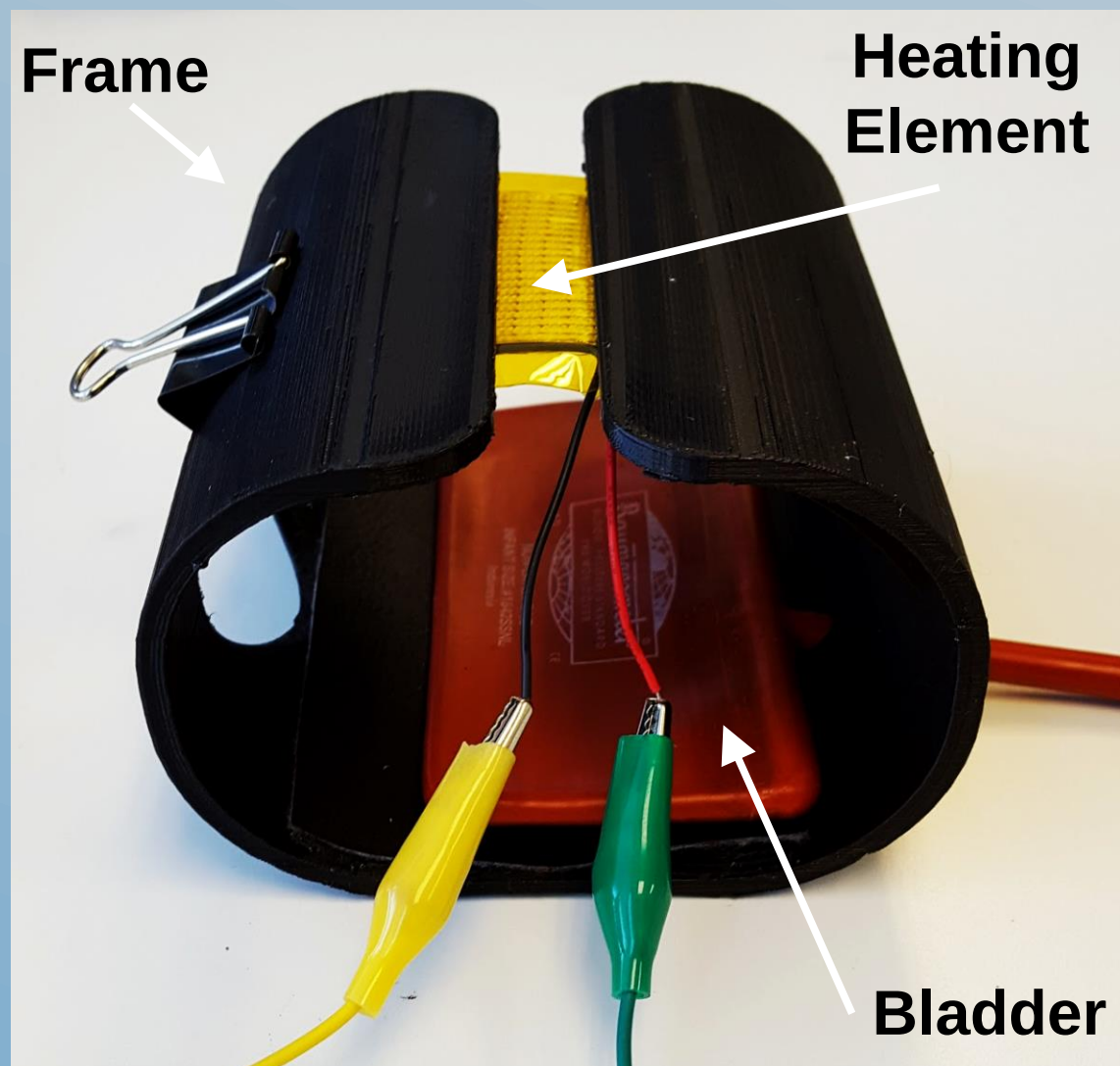


Fig. 8 - Frame System

Significance and Future Work

There is a need to design smaller, portable, easy to use, and energy efficient medical devices to reach the goal of patient-directed care. One major obstacle is poor patient compliance. The AUTO-Trac achieves the goal of self-directed care and increases compliance by automating the tedious procedures of another device whose effectiveness was already established. There are likely many opportunities for future BME Projects in devices based on already effective solutions redesigned to increase compliance.

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